Chapter 18 Notes

Section 2

Lemma 2.2

Here is why $\cup_{q_0 \in \mathcal{G}} S(q_0)$ is an M-set (and hence so is S). Let $G(x, q)$ stand for the formula

$$(\forall y \in x)(\text{Ordinal}(y) \wedge (\exists g \in R_\beta^{\mathcal{G}})(q \vDash [\![\hat{y} \approx g]\!]))$$

(this can be expanded to a real first-order formula by results in the previous chapter). If $S(q_0)$ happens to be in M, then $G(S(q_0), q_0)$ will be true in M. Now, we already know that each $S(q_0) \in M$. Then consider the following function which is definable over M:

$$x \in \mathcal{G} \wedge G(y, x)$$

Because the domain, $\mathcal{G}$, is in M, so is the range, and because we have shown each $S(q_0)$ to be in M, the range is the set of all $S(q_0)$ (for $q_0 \in \mathcal{G}$). Then the union of this, which is also in M, is just S itself.

Theorem 2.3

It is not fully clear to me why $p \vDash [\![(\text{Ordinal}(f) \wedge \text{Ordinal}(\hat{\delta})) \supset (f \in \hat{\delta} \vee \hat{\delta} \in f \vee f \approx \hat{\delta})]\!]$.

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.

Section 3

Theorem 3.2

A familiar error (see my notes on the previous chapter for what this error is) occurs in this proof.